Worksheet on Graph Application

1. The graph of $f(x)=\frac{2}{x}-2$ and $g(x)=2$ is shown.

Determine:

(1) the length of OA.

(2) the length of BC if OB = 4 units

(3) the length of OD and OE

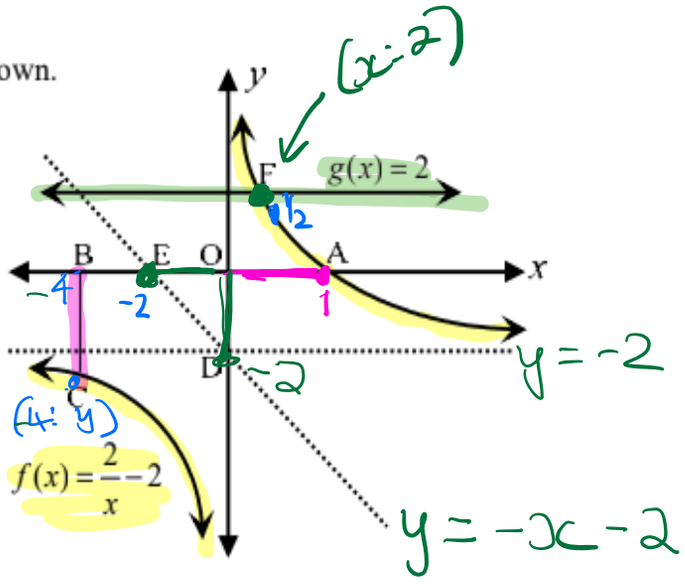
(4) the coordinates of F

(5) the values of x for which $f(x)\geq 0$

(6) the values of x for which $f(x)<0$

(7) the values of x for which $f(x)>-2$

(8) the values of x for which $f(x)>g(x)$ → graph f above graph g ?



(x: 2)

½

(4: y)

A is x-int of f

1) $0=\frac{2}{x}-2$

$\frac{2}{1}=\frac{2}{x}$

$2x=2$

$x=1$

∴ OA = 1 unit

2) Sub C (-4 : y)

$y=\frac{2}{x}-2$

$y=\frac{2}{-4}-2$

$y=-2\frac{1}{2}$

BC = 2½ units

3) D (0 : -2)

OD = 2 units

E is x-int of symmetry line

∴ $y=-x-2$

$0=-x-2$

$x=-2$

OE = 2 units

4) Sub F (x : 2) into hyp graph

$2=\frac{2}{x}-2$

$\frac{4}{1}=\frac{2}{x}$

$4x=2$

$x=1/2$

F (½ : 2)

5) $x\in(0:1]$

6) $x<0$ or $x>1$

7) $x>0$

8) $x\in(0:\frac{1}{2})$

2. The graph of $f(x)=\left(\frac{1}{2}\right)^x-2$ is shown.

Determine:

(1) the length of OA and CD.

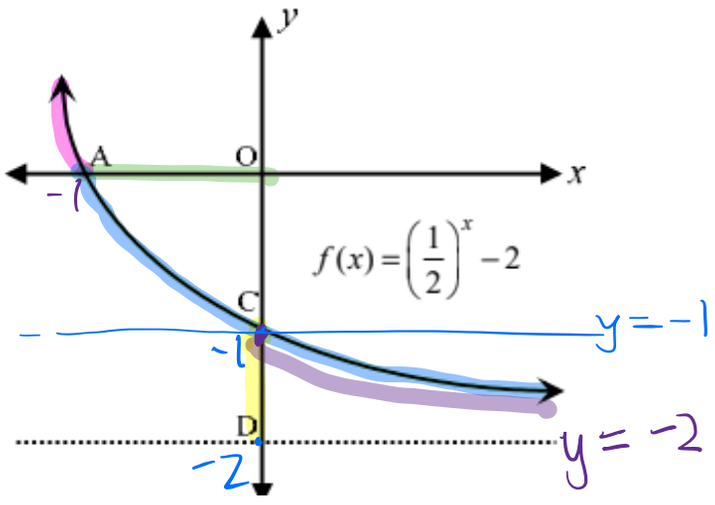
(2) the values of x for which $f(x)>0$

(3) the values of x for which $f(x)\leq 0$

(4) the values of x for which $f(x)>-2$

(5) the values of x for which $f(x)>-1$

(6) the values of x for which $f(x)\leq -1$



1) A is x-int

$0=\left(\frac{1}{2}\right)^x-2$

$2=\left(\frac{1}{2}\right)^x$

$2^1=\left(2^{-1}\right)^x$

$2^1=2^{-x}$

$-1=x$

OA = 1 unit

C is y-int

$y=\left(\frac{1}{2}\right)^0-2$

$y=1-2$

$y=-1$

∴ CD = 1 unit

2) $x<-1$

3) $x\geq -1$

4) $x\in\mathbb{R}$

5) $x<0$

6) $x\geq 0$

3.

The graph of $f(x) = a.b^x + q$ and $g(x) = \frac{a}{x} + 6$ cut the x-axis at A. The graph of f passes through the point $(-2\,;-12)$ and cuts the y-axis at 4.

Determine:

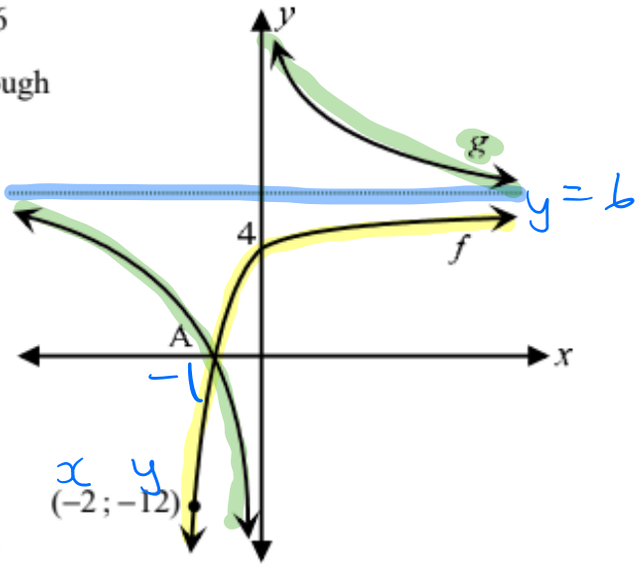
(1) the range of f

(2) the equation of f

(3) the equation of g

(4) the equation of h if the graph of f is reflected in the x-axis

(5) the values of x for which f is increasing



1) Range of f: $y \in (-\infty; 6)$

2) $y = ab^x + q$

$y = ab^x + 6$ ← asymptote

sub (0;4) – y-int

$4 = ab^0 + 6$

$-2 = a(1)$

$-2 = a$

$\therefore y = -2b^x + 6$

sub (−2; −12)

$-12 = -2b^{-2} + 6$

$-18 = -2b^{-2}$

$9 = b^{-2}$

$9 = \frac{1}{b^2}$

$9b^2 = 1$

$b^2 = \frac{1}{9}$

$b = \frac{1}{3}$ $(b > 0)$

$\therefore f(x) = -2\left(\frac{1}{3}\right)^x + 6$

3) $g(x) = \frac{a}{x} + 6$

Sub A(−1 ; 0) into graph of g

$0 = \frac{a}{-1} + 6$

$a = 6$

$\therefore \boxed{g(x) = \frac{6}{x} + 6}$

Sub A(x ; 0) into graph f

$0 = -2\left(\frac{1}{3}\right)^x + 6$

$-6 = -2\left(\frac{1}{3}\right)^x$

$3 = \left(\frac{1}{3}\right)^x$

$3 = \left(3^{-1}\right)^x$

$3^1 = 3^{-x}$

$x = -1$

4) Graph reflected in x-axis

$\therefore$ y changes sign.

$-y = -2\left(\frac{1}{3}\right)^x + 6$

$\therefore y = 2\left(\frac{1}{3}\right)^x - 6$

5) f is increasing for $x \in \mathbb{R}$

4. In the diagram below, the graphs of f and g are shown. The graphs intersect at $(1;3)$. Determine:

(1) the equation of f

(2) the coordinates of the turning point of f.

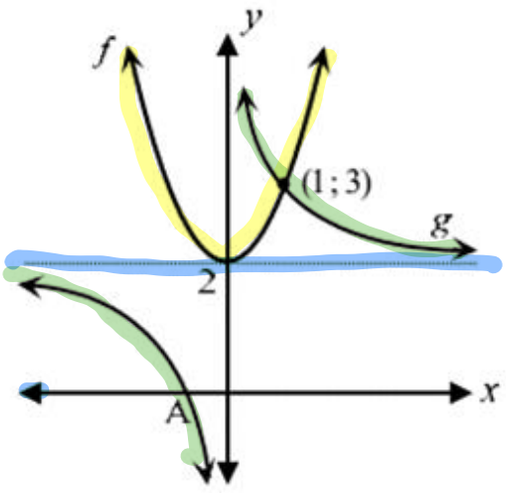
(3) the equation of the axis of symmetry of f.

(4) the equation of the horizontal asymptote of g

(5) the equation of g and the coordinates of A

(6) the domain and range of f and g

(7) the values of x for which f is decreasing?



1) $y = ax^2 + q$

$y = ax^2 + 2$ sub (1;3)

$3 = a(1)^2 + 2$

$1 = a$ $\therefore f(x) = x^2 + 2$

2) TP (0;2)

3) $x = 0$ (y-axis)

4) $y = 2$

5) $g(x) = \frac{a}{x} + q$

$y = \frac{a}{x} + 2$ sub (1;3)

$3 = \frac{a}{1} + 2$

$1 = a$

$g(x) = \frac{1}{x} + 2$

6) Graph f

Domain: $x \in R$

Range: $y \in R; y \geq 2$

Graph g

Domain: $x \in R; x \neq 0$

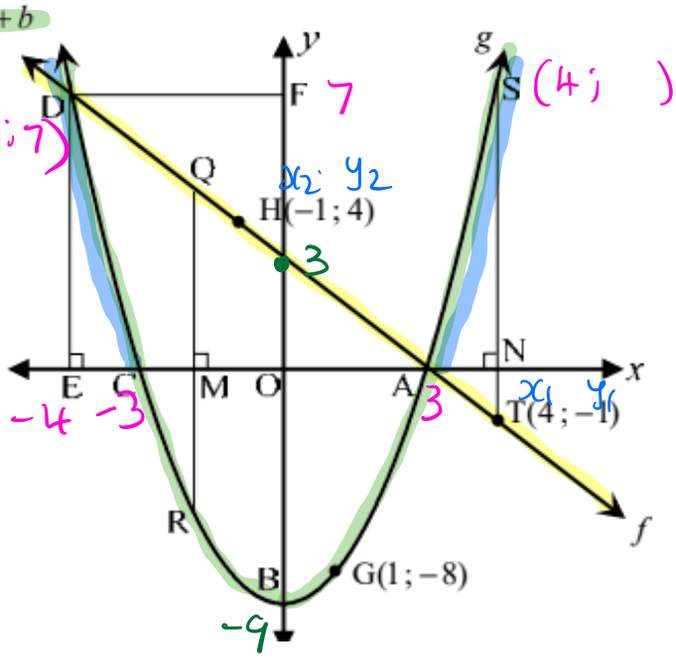
Range: $y \in R; y \neq 2$

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7) $x < 0$

5. The graph of $f(x) = mx + c$ and $g(x) = ax^2 + b$ intersect at D and on the x-axis at A.
H and T lie on line f. Determine:

(1) the equation of f
(2) the coordinates of A and hence C
(3) the equation of g
(4) the length of AC and OB
(5) the length of DE and DF
(6) the length of ST
(7) the length of OM if $QR = \frac{45}{4}$
(8) the values of x for which $g(x) \geq 0$
(9) the values of x for which $g(x) < f(x)$



1) $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - (-1)}{-1 - 4} = \frac{5}{-5} = -1$

$\therefore y = -1x + c$ sub T(4; -1)

$-1 = -1(4) + c$

$3 = c$

$\therefore \boxed{y = -1x + 3}$

2) A is x-int $\therefore$ put $y = 0$

$0 = -1x + 3$

$x = 3$

A (3; 0)
C (-3; 0)
} also x-intercepts of the parabola

3) Eqn of g: use $y = a(x - R_1)(x - R_1)$

$y = a(x - -3)(x - 3)$

$y = a(x + 3)(x - 3)$ → sub G

$-8 = a(1 + 3)(1 - 3)$

$-8 = a(4)(-2)$

$-8 = -8a$

$1 = a$

$\therefore y = 1(x + 3)(x - 3)$

$\boxed{y = x^2 - 9}$

4) AC = 6 units (use x-intercepts A & C)
OB = 9 units (use y-intercept at B)

5) To find D solve equations simultaneously:

$y_1 = (-x + 3)$ $\quad y_1 = x^2 - 9$

$\therefore -x + 3 = x^2 - 9$ (because they share "y_1")

$0 = x^2 + x - 12$

$0 = (x+4)(x-3)$

(at D) $\rightarrow$ $x = -4$ or $x = 3$ (at A)

$\therefore y = -(-4) + 3$ $\quad y = -3 + 3$

$y = 4 + 3$ $\quad y = 0$

$y = 7$ $\quad$ A (3:0)

D (-4, 7)

$\therefore$ DE = 7 units $\quad$ DF = 4 units

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6) Sub s (4; y) into parabola to find 'y'

$y = x^2 - 9$

$= 4^2 - 9$

$y = 7$ $\quad \therefore$ SN = 7 units

Given T (4; -1) $\therefore$ NT = 1 unit

$\therefore ST = SN + NT$

$= 7 + 1$

ST = 8 units

7) QR = "top graph − bottom graph"
= str line − parabola

$\frac{45}{4} = (-x+3) - (x^2 - 9)$

$\frac{45}{4} = -x + 3 - x^2 + 9$

$x^2 + x + \frac{45}{4} - 12 = 0$

$x^2 + x - 0{,}75 = 0$

$x = 1/2$ or $x = -3/2 \rightarrow M(-3/2; 0)$

$\therefore$ OM = 1 1/2 units

8) $x \leq -3$ or $x \geq 3$ see highlighted in blue

9) $x \in (-4; 3)$